

Unit-Level sUAS Training Program Planning Guide

Framework for commanders and training NCOs building organic FPV capability — without contractor dependency.

forgeandflightacademy.com | info@forgeandflightacademy.com | SAM: YV8UNYJWZHV1 / CAGE: 1A6J3

About This Guide

For unit leaders, training NCOs, and S3 shops building a sustainable organic sUAS training program. Addresses the five most common capability gaps, provides a planning framework, and maps the fastest procurement pathway.

THE CORE PROBLEM

Most units have platforms. Almost none have the training infrastructure to sustain them. The result: 60–120 day contractor waits after attrition, operators who can fly but can't repair, and zero internal capability to replicate training without outside help.

60–120

Days avg. wait on contractor support after attrition

80%

FPV losses in Ukraine attributed to EW, not enemy fire

0

Contractor dependencies after FFA graduation

Section 1: The 5 Capability Gaps

Baseline where your unit stands before selecting courses. These are the five gaps FFA sees consistently across DoD units.

1

Build Capability

Operators can fly, but cannot build or configure a platform from components. When a platform is lost or damaged, the unit is mission-stopped until contractor support arrives.

2

Field Repair

No organic repair capability. Component-level failures that take 15 minutes to fix become 60-day logistics events. Operators treat platforms as disposable because they have no choice.

3

EW Resistance

Operators have no understanding of why their links degrade, jam, or fail. They cannot adapt in real time to electronic warfare — the leading cause of FPV loss in contested environments.

4

Autonomous Systems Literacy

Operators using AI-enabled platforms cannot recognize when the system is making incorrect decisions. Human-in-the-loop oversight requires understanding how the system fails.

5

Internal Training Replication

Even when a unit gets trained, there is no certified internal instructor to replicate it. The next rotation starts from zero. Without a train-the-trainer pipeline, capability doesn't scale.

Section 2: Unit Readiness Baseline

Rate each domain 1–5. Any domain rated 1–2 is a priority training requirement.

Domain	1 — None	3 — Partial	5 — Self-Sufficient
Platform Build & Config	Cannot assemble or configure	Can configure, not build	Full build from components
Field Repair	Zero repair capability	Basic swaps only	Component-level repair
EW / RF Awareness	Cannot diagnose link issues	Recognizes degradation	Adapts in real time
Autonomous Systems	No AI/autonomy exposure	Aware of capabilities	Supervises autonomy correctly
Internal Training Pipeline	No internal instructor	Informal knowledge transfer	Certified instructor on staff

Section 3: Course Sequencing by Unit Profile

Unit Type	Start Here	Then Add	To Scale
SOCOM / SF Unit	FFF-401	FFR-201	FFF-402
Conventional Maneuver	FFF-401	FFF-301	FFF-402
R&D / Test Organization	FFA-401	FFR-201, FFF-201	FFF-402
Aviation / UAS Unit	FFR-201	FFF-401	FFF-402

Section 4: Procurement Pathway Cheat Sheet

Method	Threshold	Best For	Est. Timeline
GPC (Purchase Card)	Up to \$25K	FFR-101, FFF-301, FFA-201	Days
Simplified Acquisition	Up to \$250K	FFF-401, FFR-201 — single cohort	2–6 weeks
MIPR	No limit	Multi-course packages, large cohorts	4–12 weeks

Contracts (FAR 13/15)	No limit	FFF-402 + curriculum license + annual support	8–16 weeks
-----------------------	----------	---	------------

KEY THRESHOLD

Most single-cohort FFF-401 and FFR-201 programs fall under the \$250K simplified acquisition threshold — faster procurement, less administrative overhead. FFF-402 at 8 students (\$264,500) slightly exceeds SAT; 6-student cohorts (\$220,500) stay under for faster execution.

Section 5: Training Timeline Calculator

Objective	Course Path	Training Days	Capability Gained
Basic FPV Operator	FFF-401	10 days	Build, fly, field-repair FPV platform
EW-Resistant Operator	FFF-401 + FFR-201	15 days	Full FPV + real-time jamming response
AI-Enabled Operator	FFF-401 + FFA-401	15 days	FPV + edge AI deployment on platform
Internal Instructor	FFF-402	15 days	Certified to teach FFF-401 internally
Full Unit Capability	FFF-401 + FFR-201 + FFF-402	30 days	Self-sustaining training pipeline

Ready to Build Your Proposal?

Tell us your unit type, mission requirements, and training timeline. Proposal returned within 48 hours. SAM registered — GPC, MIPR, and simplified acquisition accepted.

forgeandflightacademy.com/contact/